
CDS Pricing

Keywords: CDS, Premium leg, loss leg, accrual

Summary: This note outlines the pricing of default swaps, which is used for both pricing single name default swap and credit curve construction using market credit default swap spreads. The pricing of default swaps consists of three parts: Premium leg, loss leg and accrual in default. We make no assumption about when default occurs even within a short time period. Instead we approximate the loss leg by using integration where the hazard rate and forward interest rate are assumed to be constant within the small discrete time period. Explicit mathematical expressions are given for each part of the pricing

1 Basic Assumption

We assume that default premium is paid at times $t_1, t_2, \dots$, and t_n , and the current time is $t_0 = 0$, and the survival time of the underlying reference asset is T whose distribution function is $F(t)$, and density function is $f(t)$. I introduce two more notations as follows: ${}_t p_x$ is the survival probability of the reference asset for another t years after it has reached to year x . It is noted that x measures one point time while t measures the duration of time¹. Default probability can be defined as follows:

$${}_t q_x = 1 - {}_t p_x.$$

Sometimes we use q_x and p_x to denote the probability of default and survival over next period of time conditional on survival to age x by omitting the length of time with an understanding of one period or one year.

2 Premium Side

Premium is usually paid at the end of each period, but it could also be paid at the beginning of the period. The condition of premium payment is the survival of the underlying reference asset.

The present value of the premium can be written as follows

$$PV(\text{premium}) = \sum_{i=1}^n \pi_i \cdot d_{t_i} \cdot {}_{t_i} p_0$$

if the premium is paid at the end of each period, and

¹Unfortunately $t_1, t_2, \dots$, and t_n are used for both the duration of time and a point of time, but is OK since we assume $t_0 = 0$

$$PV(premium) = \sum_{i=1}^n \pi_i \cdot d_{t_{i-1}} \cdot t_{i-1} p_0$$

if the premium is paid at the beginning of each period. The premium π_i is the premium for the i th period. For level premium they are a constant; for customized premium π_i they are a function of time or period.

3 Loss Payment Side

The loss side is usually assumed to be settled at the time of default even though in reality it could take quite sometime to settle the recovery value. The present value of loss side can be calculated as follows

$$PV(loss\ side) = \int_0^T N_t \cdot (1 - R_t) \cdot d_t \cdot f(t) dt$$

where d_t is the risk-free discount factor of term t . Here the recovery assumption used is an approximation to the legal treatment of corporate bond recovery in the US market at least². We assume that we recover a fixed percentage R_t of the notional amount of bond face value. Also note here that we use a continuous expression for the present value of loss without making an explicit assumption that loss occurs at a given point time within an interval. For example, we used to assume that loss occurs in the middle of a period in the previous model. In reality bankruptcy could be declared at anytime even on holidays as in the case of Worldcom. The assumption about when exactly default occurs within a period makes not too much difference for small spreads, but the difference could be substantial when spread is high as in the case of emergent market. We like to build a model to work for all ranges of spread. Both notional amount of N_t , and recovery rate R_t could be a time-varying function of time or period to accomodate the customized notional amount and recovery rate.

4 Accrual Interest over one period

If the premium is paid at the end of each period, there is an accrual payment by the protection buyer. If the premium is paid at the beginning of each period, there is a refund to the protection buyer. The present value of the adjustment at the beginning of the period can be calculated as follows over a small time interval $[0, \Delta t]$ if we assume that hazard rate and interest rate are constant over the small time interval.

$$PV(accrual\ payment) = \frac{\pi}{\Delta t} \frac{h}{r+h} \left[\frac{1}{r+h} - \left(\frac{1}{r+h} + \Delta t \right) e^{-(r+h)\Delta t} \right]$$

²For US corporate bonds, the bondholder has a claim to the face value and the accrued interest for the period when default occurs if the bond is not a zero coupon bond; For zero coupon bond the bondholder has a claim equal to the linearly interpolated value of issue price and the face value and the interpolation is based on the timing of default, issue date and maturity date.

$$PV(refund\ payment) = \frac{\pi}{\Delta t} \frac{h}{r+h} \left[\Delta t - \frac{1}{r+h} \left(1 - e^{-(r+h)\Delta t} \right) \right]$$

5 Practical implementation

Given an interest rate curve in terms of discount factors $d_1, d_2, \dots$, and d_n at times $t_1, t_2, \dots$, and t_n , and a credit curve in terms of marginal conditional default probabilities $(t_1-t_0)q_{t_0}, (t_2-t_1)q_{t_1}, \dots, (t_n-t_{n-1})q_{t_{n-1}}$ we assume that the interest rate and hazard rate over each small time interval is a constant. Then we can obtain the interest rate and hazard rate as follows:

$$r_i = -\frac{1}{\Delta t_i} \ln \left(\frac{d_i}{d_{i-1}} \right),$$

$$h_i = -\frac{1}{\Delta t_i} \ln \left(1 - (t_i - t_{i-1}) q_{t_{i-1}} \right).$$

Then we can discretize above continuous formula to get the following PV of loss and PV of accrual payment or PV of refund payment.

$$PV(loss\ side) = \sum_{i=1}^n N_{t_i} \cdot (1 - R_{t_i}) \cdot \frac{h_i}{r_i + h_i} \left(1 - e^{-(r_i+h_i)\Delta t_i} \right) \cdot d_{t_{i-1}} p_{t_0}$$

$$PV(accrual\ payment) = \sum_{i=1}^n \frac{\pi_i}{\Delta t_i} \frac{h_i}{r_i + h_i} \left[\frac{1}{r_i + h_i} - \left(\frac{1}{r_i + h_i} + \Delta t_i \right) e^{-(r_i+h_i)\Delta t_i} \right] d_{t_{i-1}} p_{t_0}$$

$$PV(refund\ payment) = \sum_{i=1}^n \frac{\pi_i}{\Delta t_i} \frac{h_i}{r_i + h_i} \left[\Delta t_i - \frac{1}{r_i + h_i} \left(1 - e^{-(r_i+h_i)\Delta t_i} \right) \right] d_{t_{i-1}} p_{t_0}$$

The break-even level premium can then be calculated using the simple accounting equality.

$$PV(loss\ side) = PV(premium) + PV(accrual\ payment)$$

or

$$PV(loss\ side) = PV(premium) - PV(refund\ payment).$$

Written by David X. Li at Columbia University on March 6, 2016. David can be reached by email: dl3054@columbia.edu.